WORKED SOLUTION

STEP III 1995, Question 2

Pure • Reduction Formulae

STEP III 1995, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s3-q2

The key idea

The factors $\sqrt{x}$ and $\sqrt{a-x}$ both suggest the substitution $x = a\sin^2\theta$, which turns them into $\sqrt{a}\sin\theta$ and $\sqrt{a}\cos\theta$ at the same time. (There is also a neat geometric shortcut for I_0.) For the reduction formula, integrate by parts so that the power of x drops by one. The power of $a-x$ then goes up by one, and we split it off using $a-x$ again. This recovers I_n itself, so we can collect terms.

Throughout, $a > 0$.

Solution

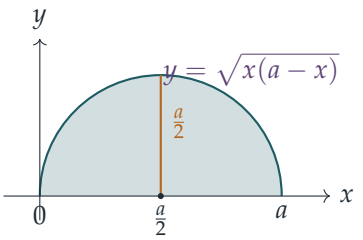
Finding I_0

We need $\mathrm{I}_0 = \int_0^a x^{1/2}(a-x)^{1/2}\,\mathrm{d}x$. Put $x = a\sin^2\theta$, so $\mathrm{d}x = 2a\sin\theta\cos\theta\,\mathrm{d}\theta$. As x runs from 0 to a, θ runs from 0 to $\frac{\pi}{2}$, and on this range $\sin\theta, \cos\theta \geq 0$. So $\sqrt{x} = \sqrt{a}\sin\theta$ and $\sqrt{a-x} = \sqrt{a}\cos\theta$, with no sign trouble. Then

$$\begin{aligned}\mathrm{I}_0 &= \int_0^{\pi/2} (\sqrt{a}\sin\theta)(\sqrt{a}\cos\theta)\,2a\sin\theta\cos\theta\,\mathrm{d}\theta = 2a^2\int_0^{\pi/2}\sin^2\theta\cos^2\theta\,\mathrm{d}\theta \\ &= \frac{a^2}{2}\int_0^{\pi/2}\sin^2 2\theta\,\mathrm{d}\theta = \frac{a^2}{4}\int_0^{\pi/2}(1-\cos 4\theta)\,\mathrm{d}\theta = \frac{a^2}{4}\cdot\frac{\pi}{2} = \frac{\pi a^2}{8},\end{aligned}$$

as required. Here we used $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$ and $\sin^2 2\theta = \frac{1}{2}(1-\cos 4\theta)$, and $\cos 4\theta$ integrates to zero over $[0, \frac{\pi}{2}]$.

A picture. If $y = \sqrt{x(a-x)}$, then $y^2 = ax - x^2$, which rearranges to $(x - \frac{a}{2})^2 + y^2 = (\frac{a}{2})^2$ with $y \geq 0$. So I_0 is the area of a semicircle of radius $\frac{a}{2}$, which is $\frac{1}{2}\pi(\frac{a}{2})^2 = \frac{\pi a^2}{8}$.



The reduction formula

Take $n \geq 1$. Integrate by parts with $u = x^{n+\frac{1}{2}}$ and $\mathrm{d}v = (a-x)^{1/2}\,\mathrm{d}x$. Then $\mathrm{d}u = (n+\frac{1}{2})x^{n-\frac{1}{2}}\,\mathrm{d}x$ and $v = -\frac{2}{3}(a-x)^{3/2}$. This gives

$$\mathrm{I}_n = \left[-\tfrac{2}{3}x^{n+\frac{1}{2}}(a-x)^{3/2}\right]_0^a + \frac{2}{3}\left(n+\frac{1}{2}\right)\int_0^a x^{n-\frac{1}{2}}(a-x)^{3/2}\,\mathrm{d}x.$$

The boundary term vanishes at both ends: at $x = a$ because of the factor $(a-x)^{3/2}$, and at $x = 0$ because of $x^{n+\frac{1}{2}}$. Note that $\frac{2}{3}(n+\frac{1}{2}) = \frac{2n+1}{3}$.

> **?** **Ask yourself** The new integral has $(a-x)^{3/2}$ in it, which is not of the form I_{n-1}. What do we do with it?

Write $(a-x)^{3/2} = (a-x)\,(a-x)^{1/2}$ and expand the first factor:

$$\int_0^a x^{n-\frac{1}{2}}(a-x)^{3/2}\,\mathrm{d}x = a\int_0^a x^{n-\frac{1}{2}}(a-x)^{1/2}\,\mathrm{d}x - \int_0^a x^{n+\frac{1}{2}}(a-x)^{1/2}\,\mathrm{d}x = a\,\mathrm{I}_{n-1} - \mathrm{I}_n.$$

So

$$\mathrm{I}_n = \frac{2n+1}{3}(a\,\mathrm{I}_{n-1} - \mathrm{I}_n) \quad\Longrightarrow\quad (2n+4)\,\mathrm{I}_n = (2n+1)\,a\,\mathrm{I}_{n-1},$$

after multiplying by 3 and collecting the I_n terms ($3 + 2n + 1 = 2n + 4$). This is the required result.

Evaluating I_n

Applying $\mathrm{I}_k = \dfrac{(2k+1)a}{2k+4}\,\mathrm{I}_{k-1}$ for $k = n, n-1, \ldots, 1$ and ending at I_0 gives

$$\mathrm{I}_n = a^n \frac{\prod_{k=1}^{n}(2k+1)}{\prod_{k=1}^{n}(2k+4)} \cdot \frac{\pi a^2}{8}.$$

Now we tidy the two products. The top is the product of the odd numbers $3 \cdot 5 \cdots (2n+1)$. We put in the missing even numbers $2 \cdot 4 \cdots (2n) = 2^n n!$ to get

$$\prod_{k=1}^{n}(2k+1) = \frac{(2n+1)!}{2^n\,n!}.$$

The bottom is $\prod_{k=1}^{n} 2(k+2) = 2^n \cdot 3 \cdot 4 \cdots (n+2) = 2^n \dfrac{(n+2)!}{2}$. Therefore

$$\mathrm{I}_n = \frac{\pi a^{n+2}}{8} \cdot \frac{(2n+1)!}{2^n\, n!} \cdot \frac{2}{2^n\,(n+2)!} = \boxed{\frac{\pi\, a^{n+2}\,(2n+1)!}{4^{n+1}\, n!\,(n+2)!}}$$

This also gives the correct value $\frac{\pi a^2}{8}$ when $n = 0$.

Checks

- $n = 1$: the recurrence gives $6\mathrm{I}_1 = 3a\mathrm{I}_0$, so $\mathrm{I}_1 = \frac{\pi a^3}{16}$. The formula gives $\pi a^3 \cdot 3!/(16 \cdot 1! \cdot 3!) = \frac{\pi a^3}{16}$, which agrees.
- Symmetry: replacing x by $a - x$ shows $\mathrm{I}_1 = \int_0^a x^{1/2}(a-x)^{3/2}\,\mathrm{d}x$ too. Adding the two expressions gives $2\mathrm{I}_1 = a\mathrm{I}_0$, which confirms $\mathrm{I}_1 = \frac{\pi a^3}{16}$ independently.
- Numerically with $a = 1$, Simpson or midpoint quadrature gives $\mathrm{I}_0 \approx 0.392699 = \pi/8$ and $\mathrm{I}_3 \approx 0.085903$. The formula gives $\pi \cdot 7!/(4^4 \cdot 3! \cdot 5!) = 35\pi/1280 \approx 0.085903$.

Common slips

- Getting the sign of v wrong: $\int (a-x)^{1/2}\,\mathrm{d}x = -\frac{2}{3}(a-x)^{3/2}$, with a minus sign from the chain rule.
- Trying to reach I_{n-1} directly instead of noticing that I_n reappears on the right and must be collected.
- Dropping the factor of 2 from $3 \cdot 4 \cdots (n+2) = (n+2)!/2$.

Beyond STEP

These integrals are special cases of the Beta function $B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,\mathrm{d}t$. Substituting $x = at$ shows $\mathrm{I}_n = a^{n+2}B\left(n+\frac{3}{2}, \frac{3}{2}\right)$. The recurrence we proved is one case of the identity $B(p+1,q) = \frac{p}{p+q}B(p,q)$. In a first-year course you meet the formula $B(p,q) = \Gamma(p)\Gamma(q)/\Gamma(p+q)$, which explains where the π comes from: $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Siklos, *Advanced Problems in Mathematics*, has more reduction-formula questions in this style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.